AIRSTREET · DESIGN DECISION RECORD

AirStreet TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	AST-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/airstreet		
Source	docs/decisions/0001-trl2-review-decisions.md @ c748953		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Item O1 decided by Amish on 2026-10-02 (AST-DEC-001)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted for TRL 3 work under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Items D1 to D9 are decided by Amish, 2026-09-25: go with recommendation (AST-DDR-002). Item O1 had no recommendation on 2026-09-25; a recommendation was written later and Amish approved it on 2026-10-02 ("i approve your recommendations for all 555 open decisions."; AST-DEC-001).

Context

The TRL 2 review note (`docs/REVIEW.md`, session 2026-09-25, /populate) listed ten items as "Proposed, awaiting Amish", nine of them with a recommendation. On 2026-09-25 Amish asked for this batch of repos to go through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Every item that carries a recommendation is therefore adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. None of them is recorded as decided or approved by Amish. The item without a recommendation stays open.

Options considered

The options for each item are those listed in `docs/REVIEW.md` (session 2026-09-25, /populate) and in AST-PRC-001 v0.2, Key design choices.

Decision

Table 1. Items decided (v0.1 recorded them as adopted for TRL 3 work, open for review; accepted by Amish on 2026-09-25).

#	Item	Recommendation adopted	Status
D1	Budget	Option (a): `budget_usd` covers the AirStreet sensor head only; the FieldNode core is costed in the FieldNode repo. R15 is redefined as the sensor-head cost. The figure itself was set at $285 under AST-DDR-002 (the $280 recommended here was overtaken by the TRL 3 price of $283.00).	Decided by Amish, 2026-09-25: go with recommendation
D2	Pitch wording	"A street-level air quality node measuring PM2.5 and NO2, with particles checked on CalRig and NO2 calibrated against a reference station, for neighborhood-scale pollution maps." Applied to `project.yaml` and `README.md`.	Decided by Amish, 2026-09-25: go with recommendation
D3	NO2 calibration method	Field collocation with a regulatory reference station for at least 14 days before deployment and at least every 6 months after, with one permanently collocated anchor node; multiple linear regression first, random forest later.	Decided by Amish, 2026-09-25: go with recommendation
D4	NO2 sensor	Alphasense NO2-B43F class (four electrodes, ozone filter), not a three-electrode sensor.	Decided by Amish, 2026-09-25: go with recommendation
D5	NO2 front end	Buy an ISB class board for the first units; the open two-channel potentiostat stays a later cost option.	Decided by Amish, 2026-09-25: go with recommendation
D6	PM sensor	Sensirion SPS30 class, not a Plantower PMS5003 class sensor.	Decided by Amish, 2026-09-25: go with recommendation
D7	Inlet height	About 3.0 m above the sidewalk, within the EU 1.5 to 4 m range.	Decided by Amish, 2026-09-25: go with recommendation
D8	Reporting	One record every 5 min, PM sensor run 30 s per record, raw signals sent.	Decided by Amish, 2026-09-25: go with recommendation
D9	Open data	Publish calibrated hourly data, with the calibration version, to an open platform such as OpenAQ, subject to the partner's agreement.	Decided by Amish, 2026-09-25: go with recommendation

Table 2. Items that remain open.

#	Item	Status
O1	First partner, city and reference station for co-design and collocation. No recommendation was made.	Decided by Amish, 2026-10-02: a city or air agency with a street-level regulatory NO2 reference station that will share its data; first candidate to approach, TCEQ's Dallas-Fort Worth monitoring network with a local university or city partner (AST-DEC-001)

Consequences

- `project.yaml` : pitch replaced per D2; `budget_usd` set to $285 under AST-DDR-002 (D1 fixed its scope, the sensor head). The problem line is unchanged, since no rewording was recommended.

- AST-REQ-001 v0.3: R15 now covers the sensor head only, with the FieldNode core excluded and costed in FieldNode (D1). Targets for R6, R8 and R11 are unchanged but now rest on adopted rather than proposed choices (D3, D7, D8).
- AST-PRC-001 v0.3 and AST-PRB-001 v0.3: the design choices above are no longer shown as proposed; they are adopted for TRL 3 work, open for Amish's review.
- AST-CAL-001 v0.2 states the sensor-head cost, $285.00, against the $285 `budget_usd`.
- The TRL 3 calculations raised further items; Amish accepted their recommendations on 2026-09-25, and they are recorded in AST-DDR-002.
- Purchasing (D5), field collocation (D3) and data publication (D9) are TRL 4 and later work and are on hold by Amish's instruction.